

6: Logistical Regression!

question 8

$$P = \frac{1}{1 + e^{-(3 + 0.6 \times 3)}} =$$

$$\frac{1}{1 + e^{-(1.2)}} = \frac{1}{1 + e^{1.2}} = \frac{1}{1 + 3.32} = \underline{0.231}$$

the probability that an email with 3 suspicious keywords is spam is 23%

question 9

$$P = \frac{1}{1 + e^{-(3 + 0.6 \times 8)}} = \frac{1}{1 + e^{-(3 + 4.8)}} =$$

$$\frac{1}{1 + e^{1.8}} = \frac{1}{1 + 0.165} = \underline{0.858}$$

the probability here is 85.8%

Bonus question:

$$X_{(0.8)} = \frac{\ln\left(\frac{0.8}{1-0.8}\right) - (-3)}{0.6} =$$

$$= \frac{\ln(4) + 3}{0.6} = \frac{1.386 + 3}{0.6}$$

X = 7.31 is the number of

suspicious keywords at which the filter is 80% sure the email is spam